

Optimal Block Time for AMM Liquidity Providers under Jump-Diffusion Prices

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The case for faster blocks

Two live conversations about how fast a chain should be

Arbitrum Research — “The power of faster blocks” (Felten, May 2024)

“Arbitrum’s 250 millisecond block time leads to **65% lower arbitrage loss** compared to a 2 second block time.”

research.arbitrum.io/t/the-power-of-faster-blocks/9609 ↗

EIP-7782 “Reduce Block Latency” — 12 s → 6 s, drafted for Glamsterdam

“More frequent blocks **decrease LVR** (Loss Versus Rebalancing), which improves the economics for liquidity providers. More liquidity means lower spreads.”

eips.ethereum.org/EIPS/eip-7782 ↗

Both rest on the same result. Both, on the numbers below, overstate the gain — by roughly a factor of two.

Where that argument comes from

Passive liquidity providers on an AMM lose to arbitrageurs whenever the external price moves — **loss-versus-rebalancing** (LVR), Milionis et al. (2022).

- Under geometric Brownian motion, a constant-product pool loses at the rate $\sigma^2 V/8$, independent of the block time.
- Add a swap fee γ and Poisson blocks of mean Δt (Milionis et al., 2023): the rate is multiplied by $F(\kappa)$, $\kappa = \gamma/(\sigma\sqrt{\Delta t})$ — the stationary probability that an arriving block offers a *profitable* trade.
- $F(\kappa) \sim \sigma\sqrt{\Delta t}/(\sqrt{2}\gamma) \rightarrow 0$ as $\Delta t \rightarrow 0$. That $\sqrt{\Delta t}$ is where Arbitrum's 65% comes from, and the $\sim 29\%$ implied by halving Ethereum's slot; and it says shorter blocks mean less LVR, all the way down.

The deployed prescription

“Shorten blocks.” Solana at 400 ms, Ethereum L2s at 2 s, app-chains chasing 50 ms — with LP welfare among the stated reasons.

The primitive that prescription assumes

The whole argument rests on the reference price being **continuous**. A diffusive gap has size $O(\sigma\sqrt{\Delta t})$, so halving the block halves what the arbitrageur finds.

Crypto prices are not continuous:

- Scaillet et al. (2020): frequent, clustered jumps in BTC at high frequency.
- Saef et al. (2024): BTC jumps on $\sim 58\%$, ETH on $\sim 32\%$ of trading days.
- My own ETH/USDT calibration: $\lambda = 283$ jumps/yr.

A jump does not care about Δt

A jump of size J opens a price gap of size J whatever the block schedule. Faster blocks clear each gap *sooner* — they do not make it *smaller*.

Contributions

1. An **additive LVR decomposition** under any finite-activity jump law: the classical diffusion term $\sigma^2 V/8$ plus a jump term carrying no Δt (Theorem 1).
2. A **separated rate** with fees and discrete blocks, $\ell_0 = \frac{\sigma^2 V}{8} F(\kappa) + \lambda VG_\nu(\gamma)$ — the two channels meeting only in a remainder bounded on both sides, with no order symbols (Theorem 2, Proposition 1).
3. An **exact jump floor**: $\ell(\Delta t) \geq \lambda VG_\nu(\gamma) > 0$ at every Δt for symmetric ν , with no remainder, and approached only as $\sqrt{\Delta t}$ (Theorem 3, Corollary 1).
4. A **closed-form planner's optimum** — a cubic, invariant in pool size exactly and in (λ, m, δ) below the flatness of the objective (Theorem 4), calibrated to ETH/USDT, 2020–2026.

Headline

The block schedule governs one channel and cannot touch the other. At Ethereum's 12 s that is 471 bp/yr against an exact floor of 125, and an LP-side optimum near 8 s — so the two proposals above are worth **35%** and **14%**, not 65% and 29%.

Two channels, and a floor

Setup

Milionis et al. with **one primitive changed**: the reference price jumps.

$$\frac{dS_t}{S_{t-}} = \mu dt + \sigma dW_t + (e^{J_t} - 1) dN_t, \quad N \text{ Poisson}(\lambda), \quad J \sim \nu.$$

- ν finite activity with $\int (e^{j/2} - 1)^2 \nu(dj) < \infty$; Merton $J \sim \mathcal{N}(m, \delta^2)$ the running instance.
- CPMM, $x_t y_t = L$, pool value $V(P) = 2\sqrt{LP}$ — concave, and that concavity is the loss.
- Fee γ in log-price units $\Rightarrow$ no-arbitrage band $\{|\log S - \log P| \leq \gamma\}$; blocks Poisson of rate $1/\Delta t$.
- $\text{LVR}(T) = \int_0^T V'(P_{s-}) dP_s - [V(P_T) - V(P_0)]$; rate $\ell = \mathbb{E}_t[d\text{LVR}_t]/dt$.

What is general and what is not

Results need **symmetry** of ν , not Gaussianity. Theorems 1, 2, 4 hold for arbitrary ν ; Proposition 1 and Theorem 3 need symmetry.

Theorem 1 — the zero-free decomposition

In the continuous-arbitrage limit ($\gamma = 0$, $\Delta t \rightarrow 0$), $P_t \equiv S_t$ and LVR is the Itô–Lévy remainder of $V(S_t)$:

$$\ell = \underbrace{\frac{\sigma^2}{8} V(S_t)}_{\text{diffusion-LVR}} + \underbrace{\frac{\lambda}{2} V(S_t) \mathbb{E}[(e^{J/2} - 1)^2]}_{\text{jump-LVR}},$$

and for Merton $\mathbb{E}[(e^{J/2} - 1)^2] = e^{m+\delta^2/2} - 2e^{m/2+\delta^2/8} + 1 \geq 0$, zero iff $J \equiv 0$.

Two observations

The jump term carries **no** Δt . At $\gamma = 0$ the arbitrageur clears any gap in the next block, so refining the schedule only redistributes *when* the loss lands.

Measure

The drift cancels between the two terms, so no normalisation of μ is needed — but with jumps the rate is *not* measure-independent, so this is a physical-measure statement and calibration is against realised prices.

The mispricing process

Theorem 1 had $\gamma = 0$, so the pool tracked the reference exactly. Switch the fee on and a **state** appears between them: $z_t := \log(S_t/P_t)$.

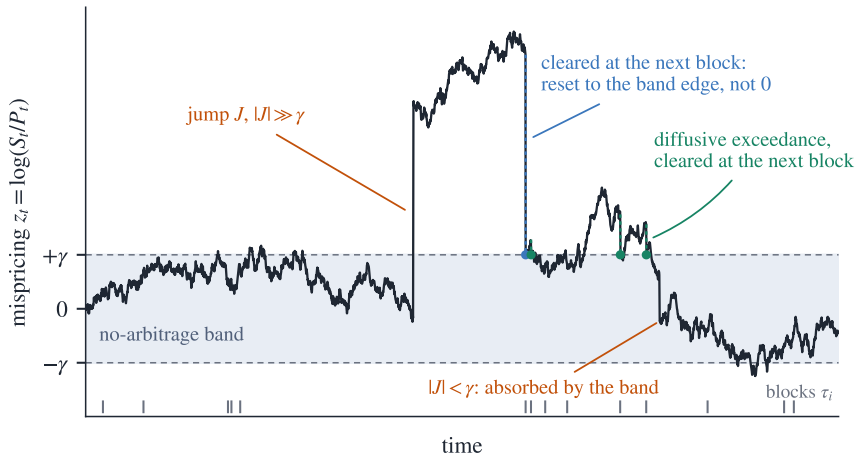
- Between blocks P is frozen, so z inherits the dynamics of $\log S$ — diffusion *and* jumps.
- At a block the arbitrageur trades iff $|z| > \gamma$, and moves the pool to the *near band edge*: $z \mapsto \text{bound}(z, -\gamma, \gamma)$ — **not to 0**. The mispricing persists.

$$z_t = \left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t + \underbrace{\sum_{k: T_k \leq t} J_k}_{\text{exogenous, rate } \lambda} + \underbrace{\sum_{i: \tau_i \leq t} \mathcal{J}_i}_{\text{endogenous, rate } 1/\Delta t}$$

The asymmetry every result comes from

Two point processes drive z : price jumps push it **away** from the band at a rate the protocol does not choose; block-triggered resets push it **back** at a rate the protocol does choose.

The mispricing process, drawn



Between blocks z diffuses and jumps; each block (ticks) resets it to $\pm\gamma$ — teal a diffusive exceedance, orange a jump.

From the process to a rate

A block with pre-trade mispricing z moves the pool's log-price by $\xi = \text{sign}(z)(|z| - \gamma)_+$, and by the computation behind Theorem 1 the LP's loss is $h(\xi) = \frac{V}{2}(e^{\xi/2} - 1)^2$. Blocks are Poisson and independent of z , so the pre-block law is the stationary law π and

$$\ell(\Delta t) = \frac{1}{\Delta t} \mathbb{E}_\pi[h(\xi)], \quad h(\xi) = \underbrace{\frac{V}{8}\xi^2}_{\text{quadratic surrogate}} + \frac{V}{2}\rho(\xi), \quad \rho(\xi) := (e^{\xi/2} - 1)^2 - \frac{\xi^2}{4}.$$

Everything below is read off this one expression, and two exact facts about the split carry it:

- $\frac{V}{8}\xi^2$ is **convex** in z , which h itself is not — a downward move of any size costs at most $V/2$. Convexity is what the floor (Theorem 3) needs.
- ρ is **signed** under symmetry: $\frac{1}{2}[\rho(\xi) + \rho(-\xi)] = \sum_{n \geq 2} \frac{1-2^{1-2n}}{(2n)!} \xi^{2n} \geq 0$, leading term $\frac{7}{192}\xi^4$. So the surrogate *understates* the loss, and the residual is carried exactly, never discarded.

The two channels, separately

Diffusion (Lemma 1). At $\lambda = 0$ the stationary law is uniform on the band with exponential tails, so

$$\ell_{\text{diff}} = \frac{\sigma^2}{8} V \cdot F(\kappa), \quad F(\kappa) = \frac{1}{1 + \sqrt{2} \kappa}$$

with $\kappa = \gamma/(\sigma\sqrt{\Delta t})$; $F(\kappa) = \mathbb{P}_\pi(|z| > \gamma)$, strictly decreasing, $F(\kappa) \rightarrow 0$.

Jumps (Lemma 2). The jump contribution is $\lambda V G_\nu(\gamma)$ with

$$G_\nu(\gamma) = \frac{1}{8} \mathbb{E}_\nu[(|J| - \gamma)^2 \mathbf{1}\{|J| > \gamma\}],$$

a functional of ν alone. For symmetric Merton, $G = \frac{\delta^2}{8} \Psi(\gamma/\delta)$.

The fee discount is different on each channel

Both channels are trimmed by the fee — but by $F(\gamma/\sigma\sqrt{\Delta t})$ on one and by $\Psi(\gamma/\delta)$ on the other.

Only the first contains Δt , because the size of a jump is set by δ , not by how long the block lasted.

Theorem 2 — the separated rate

$$\ell(\Delta t) = \ell_0(\Delta t) + \mathcal{E}(\Delta t), \quad \ell_0(\Delta t) := \frac{\sigma^2}{8} V \cdot F\left(\frac{\gamma}{\sigma\sqrt{\Delta t}}\right) + \lambda V \cdot G_\nu(\gamma).$$

This is a *definition* of $\mathcal{E}$ and carries no content until $\mathcal{E}$ is controlled.

Proposition 1 bounds it on both sides — six signed terms, no order symbols, under a stated mixing condition $n_* \leq 2(1 + \eta^2)$ checked numerically over every block time, volatility and fee tier considered:

- interaction ($|M| \leq \gamma$ pathwise), block clock, single-jump thinning, multi-jump, curvature residual (exact, closed form), stationary law.
- At the calibration: $\mathcal{E}(\Delta t) \in [-0.653, +0.849]$ bp/yr at 12 s, never above 0.29% of the rate over 50 ms–12 s. Quadrature puts the realised remainder at +0.165 bp/yr, inside the bound and of the predicted sign.

Theorem 3 — an exact floor

No approximation, no remainder

Let ν be symmetric and satisfy $\mathbb{E}[g(\sum_1^k J)] \geq k \mathbb{E}[g(J)]$ for every k (Merton with $m = 0$ does).
Then for every $\Delta t > 0$

$$\ell(\Delta t) \geq \lambda V G_\nu(\gamma) = \inf_{\Delta t' > 0} \ell_0(\Delta t') > 0.$$

- Proved from convexity of the quadratic surrogate — it never touches the stationary law, which for $\lambda > 0$ has no closed form. That is what makes it available, and why it does *not* inherit Proposition 1's mixing condition.
- **Corollary 1.** ℓ_0 is strictly increasing and the floor is approached only as $\sqrt{\Delta t}$: halving the block time removes at most 29% of the diffusion term.

Keep two limits apart

$\Delta t \rightarrow 0$ shortens the interval between blocks. The jump law is exogenous to the schedule, so it does *not* shrink J .

Numbers, and the planner's optimum

Calibration, and the rate across deployed block times

Binance ETH/USDT 5-minute closes, Jan 2020 – Jun 2026 (2,373 days, 682,946 bars): $\sigma = 0.8156$ from bipower variation; $\lambda = 283.3/\text{yr}$ and $\delta = 0.0192$ from Lee–Mykland at $\alpha = 1\%$; $m = 0$ ($\hat{m} = -0.0012$, 99% CI contains zero). Modelling choices: $\gamma = 5$ bp (Uniswap 0.05%), $V = \$1\text{M}$.

Chain	Δt	ℓ (bp/yr)	diff. share
Ethereum L1	12 s	471	73%
Base / OP L2	2 s	312	60%
Solana	400 ms	221	43%
Arbitrum	250 ms	203	38%
App-chain	50 ms	162	23%
Jump floor	—	125	0%

- Ceiling $\sigma^2 V/8 = 832$ bp/yr; floor $\lambda VG = 125$.
- A $240\times$ cut, $12\text{ s} \rightarrow 50\text{ ms}$, removes only 89% of the diffusion term and leaves the rate 29% above the floor.
- The channels are equal at $\Delta t_{\times} \approx 0.75\text{ s}$; the diffusion term reaches a tenth of the floor only at $\Delta t \approx 5.5\text{ ms}$.

Theorem 4 — netting LVR against what a block costs

Fixed per-block work gives a cost rate $R(\Delta t) = r/\Delta t$; a pool inherits $c/\Delta t$ with $c = r V/V_{\text{chain}}$. The LP-side planner minimises $W(\Delta t) = \ell_0(\Delta t) + c/\Delta t$:

$$\eta^{\text{opt}}(1 + \eta^{\text{opt}})^2 = \frac{V\gamma^2}{8c}, \quad \Delta t^{\text{opt}} = \frac{2\gamma^2}{\sigma^2(\eta^{\text{opt}})^2},$$

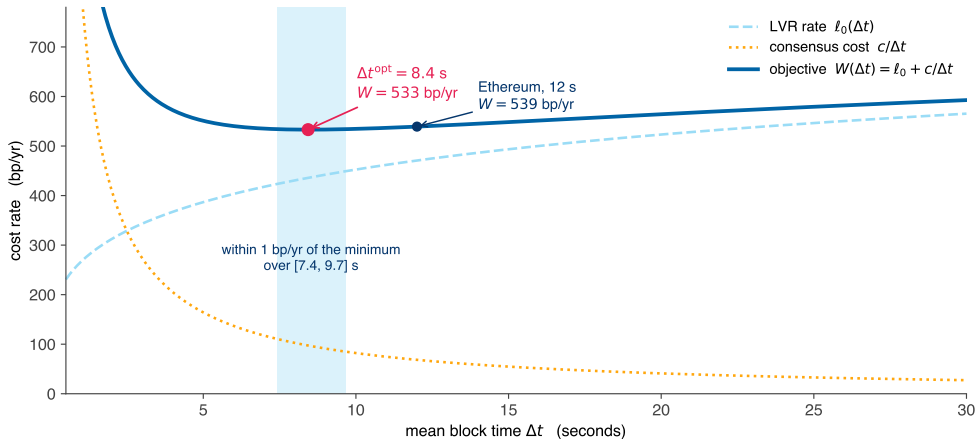
a cubic with a unique positive root, closed form by Cardano.

- $\partial\Delta t^{\text{opt}}/\partial\lambda = \partial\Delta t^{\text{opt}}/\partial m = \partial\Delta t^{\text{opt}}/\partial\delta = 0$: **the jump parameters do not enter the first-order condition.**
- $\partial\Delta t^{\text{opt}}/\partial V = 0$, exactly — V cancels, since $c \propto V$. The optimum is a property of the chain and the fee tier, not of the pool.

What the remainder does to it

That invariance is a property of ℓ_0 ; $\mathcal{E}$ carries both λ and Δt . Bounding it directly puts any exact minimiser in [7.3, 9.9] s — the range over which W is within 1 bp/yr of its minimum anyway.

The objective, and its shallow optimum



The optimum is shallow: 539 $\rightarrow$ 533 bp/yr. Across plausible c it runs 0.2–15 s; the jump parameters move it not at all.

What this means, and what it does not

Policy

Back to the two proposals

Arbitrum, 2 s $\rightarrow$ 250 ms: $\sqrt{\Delta t}$ gives 65%, the model gives 312 $\rightarrow$ 203 bp/yr, a 35% cut.

EIP-7782, 12 s $\rightarrow$ 6 s: $\sqrt{\Delta t}$ gives 29%, the model gives 471 $\rightarrow$ 403 bp/yr, a 14% cut. Both price the diffusion channel as the whole loss.

- Shortening blocks can at most close $\ell(\Delta t) - \lambda VG$ — and only as $\sqrt{\Delta t}$.
- The diffusion channel stays material into the low-millisecond range. **So at every deployed block time, what limits block-time policy is not the floor — it is the cost of consensus.**
- Once the floor does bind, the remaining LP-side levers are not the schedule:
 - ▶ fee-tier design, since the binding ratio is γ/δ ;
 - ▶ batch auctions, which remove the per-block arbitrage window at source;
 - ▶ LVR-rebate designs, e.g. auction-managed AMMs routing pool-management revenue back to LPs.

Key takeaway

Jumps change what block time can *achieve*, not what it should *be*.

Conclusion

- **125 bp/yr** — the jump floor. Exact for symmetric ν , at every block time, with no remainder.
- **471 bp/yr** at Ethereum's 12 s, 73% of it in the channel the schedule governs — and that channel yields only as $\sqrt{\Delta t}$.
- **8.4 s** — the LP-side optimum. A cubic; invariant in pool size exactly, and in (λ, m, δ) below the objective's flatness.
- So both proposals I opened with are right in direction, and about twice too optimistic.

Key takeaway

The rate splits into a channel the block schedule governs and one it cannot touch. The floor is genuine but distant: what binds at deployed block times is the cost of a block, and the lever against the floor is γ/δ .

This is an LP-side input to a larger welfare problem, not a prescription — limitations on the backup slides. Extended version, with all proofs: [arXiv:2608.30321](https://arxiv.org/abs/2608.30321) ↗.

Questions and discussion.

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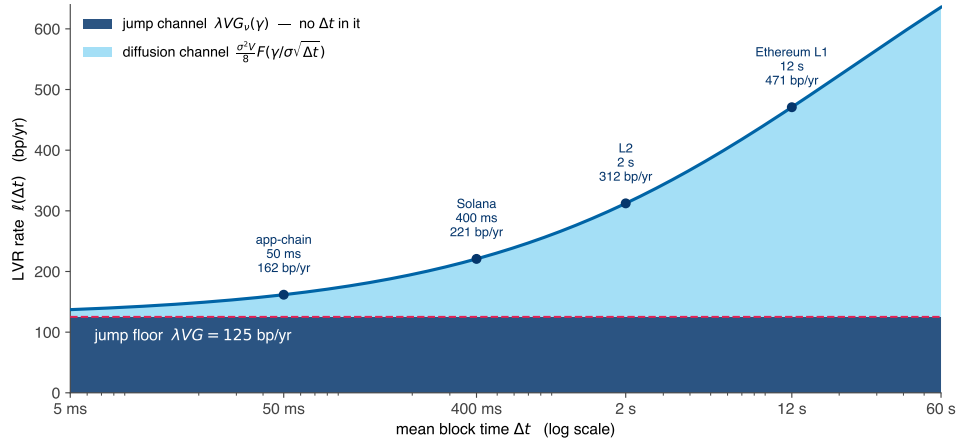
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Backup

Limitations

- **Objective is LP-LVR-only.** A full welfare treatment must also weigh MEV redistribution, finality, censorship resistance and oracle freshness — all of which could push Δt^{opt} back up.
- **Consensus cost.** Issuance is a chosen compensation policy, not a measured resource cost; it overstates where it subsidises growth and understates where MEV and fees fund validators.
- **Full-range pool.** Most LVR is paid on concentrated liquidity, where the two effects on G run in opposite directions — so the full-range floor is not simply a lower bound there.
- **Idealised arbitrageur:** unlimited capital, zero latency, continuous sight of S_t — weakest exactly in the sub-second regime the floor argument concerns.
- **Poisson blocks**, inherited for tractability. Deterministic slots give the same shape with $C = -\zeta(1/2)/\sqrt{2\pi} \approx 0.583$ in place of $1/\sqrt{2}$, and move Δt^{opt} by 5%.

The two channels, drawn



$\ell_0(\Delta t)$ at the ETH/USDT calibration. The dark band is constant in Δt ; only the pale band responds to the block schedule.

Fees split LVR — they do not reduce it

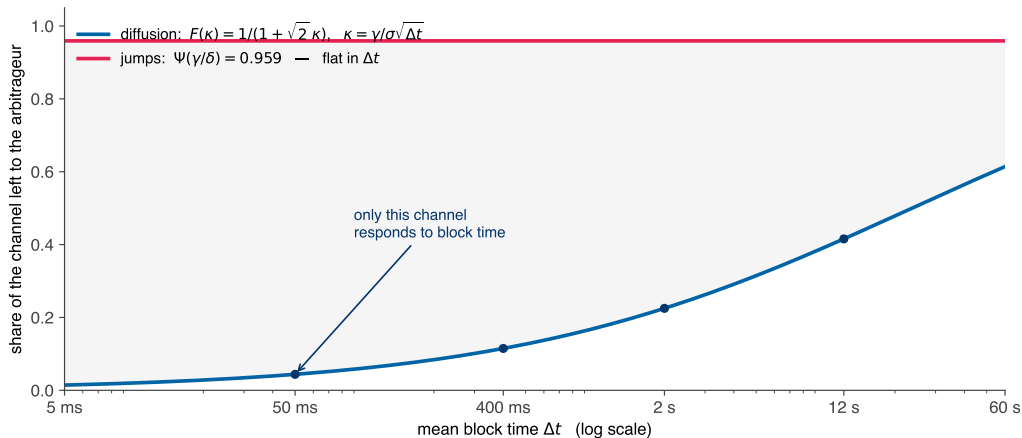
In the accounting of Milionis et al., arbitrage profit and fee income sum to the frictionless LVR of Theorem 1. So $\ell(\Delta t)$ is the arbitrage half: LP loss *net* of the fees LPs get back.

- Diffusion channel: the split is $F(\kappa) : 1 - F(\kappa)$, and $F(\kappa) \rightarrow 0$. Shorter blocks *do* return diffusion-LVR to LPs as fee income.
- Jump channel: the split is $\Psi(\gamma/\delta) : 1 - \Psi(\gamma/\delta)$ — a function of γ/δ alone, with no Δt in it.
- At the calibration $\gamma/\delta = 0.026$, so fees recover 4.1% of jump-LVR. The rest is the floor.

Key takeaway

The lever against the floor is the ratio of fee to jump size — fee-tier design — not the block schedule.

Two fee discounts, one lever



The share of each channel left to the arbitrageur. F collapses as blocks shorten; $\Psi(\gamma/\delta)$ does not move.

Which regime a chain is in matters more than its slot

ℓ in bp/yr at $\Delta t = 12$ s, diffusion share in brackets:

$\sigma \setminus \lambda$	71/yr	283/yr	1,133/yr	4,533/yr
0.30 (calm)	55 (43%)	148 (16%)	524 (4%)	2,027 (1%)
0.816 (baseline)	377 (92%)	471 (73%)	847 (41%)	2,349 (15%)
1.50 (stress)	1,625 (98%)	1,719 (93%)	2,095 (76%)	3,597 (44%)

- The *level* spans a factor of 65; the *share* runs from 1% to 98% and moves the *other* way in λ .
- The share is what decides whether shortening blocks helps at all. At $\lambda = 283/\text{yr}$ it crosses 50% at $\sigma = 0.55$.

Key takeaway

Any policy claim about 12-second blocks is conditional on the volatility-and-jump-intensity regime, not on a single point estimate.

What actually moves the optimum

$c (\times 10^{-5} \text{ USD/block})$	Δt^{opt}
500	15.4 s
260 (baseline)	8.4 s
90	3.4 s
30	1.4 s
2	0.20 s

σ	Δt^{opt}
0.30 (calm)	62.3 s
0.816	8.4 s
1.50 (stress)	2.5 s

- Nearly two orders of magnitude across the plausible range of the cost calibration; volatility moves it comparably, since $\Delta t^{\text{opt}} \propto \gamma^2 / \sigma^2$ at fixed η .
- The jump parameters do not move it at all.

Key takeaway

Under an LP-LVR-only objective, Ethereum's 12 s is somewhat too slow and sub-second targets overshoot — but the gain over the status quo is under 2%, and the answer is driven by c , σ , γ .

The reset-to-zero shortcut

A common simplification is to assume a trade resets the mispricing z to 0 rather than to the band edge $\pm\gamma$ — i.e. to use $\Psi(\kappa)$ in place of $P_{\text{trade}} = F(\kappa)$.

It is not a small error

It understates the diffusion term by $2.7\times$ at 12 s, $76\times$ at 2 s, and 4×10^7 at 400 ms — because Ψ has a Gaussian tail where F decays only algebraically.

The persistence of the mispricing (Lemma 1) is what makes the diffusion channel decay as $\sqrt{\Delta t}$ rather than vanish, and so it is also what keeps that channel material down into the millisecond range.